

Detection Logic

Overview

This project follows a log-based detection approach using Sysmon and Splunk.

Instead of relying on signatures, detection is based on analyzing system behavior through logs.

All activities are captured using Sysmon and analyzed in Splunk using SPL (Search Processing Language) queries.

Detection Strategy

The detection logic is based on:

Monitoring process execution

Analyzing command-line activity

Tracking network connections

Identifying abnormal behavior patterns

 Focus: Behavior-based detection

Log Sources

Source Description

Sysmon Event ID 1	Process Creation
Sysmon Event ID 3	Network Connections
Sysmon Event ID 13	Registry Changes

Detection Rules

1. PowerShell Execution Detection

index=main EventCode=1 CommandLine="*powershell*"

 Purpose:

Detects PowerShell usage, commonly used in attacks.

 Why Suspicious:

Attackers often use PowerShell to execute malicious scripts.

 MITRE ATT&CK:

T1059 – Command and Scripting Interpreter

2. Encoded Command Detection

index=main EventCode=1 CommandLine="*-enc*" OR CommandLine="*EncodedCommand*"

 Purpose:

Detects obfuscated PowerShell commands.

 Why Suspicious:

Attackers encode commands to bypass detection.

 MITRE:

T1027 – Obfuscated Files or Information

3. Command Prompt Activity

index=main EventCode=1 CommandLine="*cmd*"

 Purpose:

Detects command prompt usage.

 Why Suspicious:

Used for executing system commands.

4. Network Activity Detection

index=main EventCode=3

 Purpose:

Detects outbound network connections.

 Why Suspicious:

May indicate communication with external servers (C2).

 MITRE:

T1071 – Application Layer Protocol

5. Registry Persistence Detection

index=main EventCode=13

 Purpose:

Detects registry changes.

 Why Suspicious:

Attackers use registry keys for persistence.

 MITRE:

T1547 – Boot or Logon Autostart Execution

 6. Suspicious Parent-Child Process

index=main EventCode=1 ParentImage="*cmd.exe" Image="*powershell.exe"

 Purpose:

Detects unusual process relationships.

 Why Suspicious:

CMD launching PowerShell may indicate malicious activity.

 Detection Workflow

Command Execution → Sysmon Log → Forwarder → Splunk Index

→ Detection Query → Alert → Dashboard Visualization

 Visualization & Alerting

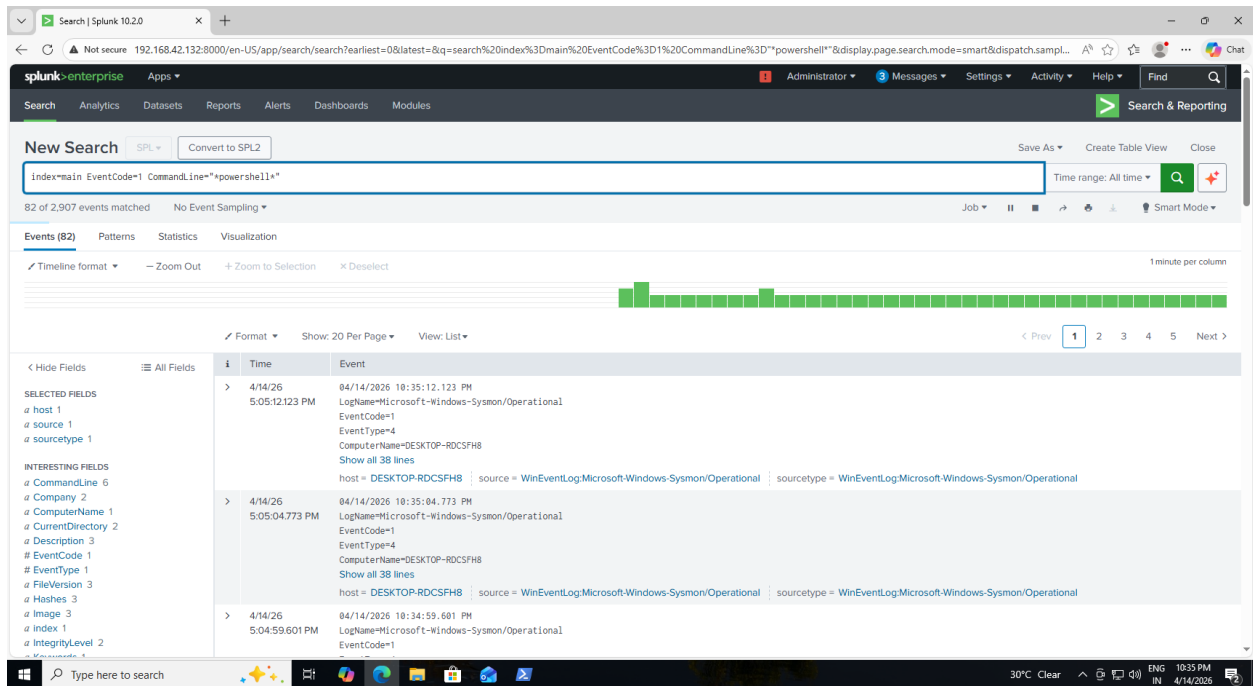
Detection rules are used to:

Trigger alerts for suspicious behavior

Populate dashboard panels

Provide real-time monitoring

Screenshot Example



The screenshot shows the Splunk Enterprise web interface. The search bar contains the query: `index=main EventCode=1 CommandLine="powershell"`. The search results show 82 events. The interface includes a sidebar with field lists, a main search results table, and a top navigation bar with options like Search, Analytics, Datasets, Reports, Alerts, Dashboards, and Modules.

New Search

82 of 2,907 events matched No Event Sampling

Events (82) Patterns Statistics Visualization

Timeline format Zoom Out Zoom to Selection Deselect

Format Show: 20 Per Page View: List

Time	Event
4/14/26 5:05:12.123 PM	04/14/2026 10:35:12.123 PM LogName=Microsoft-Windows-Sysmon/Operational EventCode=1 EventType=4 ComputerName=DESKTOP-RDCSFH8 Show all 38 lines host = DESKTOP-RDCSFH8 source = WinEventLog:Microsoft-Windows-Sysmon/Operational sourcetype = WinEventLog:Microsoft-Windows-Sysmon/Operational
4/14/26 5:05:04.773 PM	04/14/2026 10:35:04.773 PM LogName=Microsoft-Windows-Sysmon/Operational EventCode=1 EventType=4 ComputerName=DESKTOP-RDCSFH8 Show all 38 lines host = DESKTOP-RDCSFH8 source = WinEventLog:Microsoft-Windows-Sysmon/Operational sourcetype = WinEventLog:Microsoft-Windows-Sysmon/Operational
4/14/26 5:04:59.601 PM	04/14/2026 10:34:59.601 PM LogName=Microsoft-Windows-Sysmon/Operational EventCode=1

Key Takeaways

Detection is based on behavior, not signatures

Sysmon provides deep system visibility

Splunk enables powerful log analysis

Combining logs + queries = effective threat detection